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Journal of
Multimedia Technology &
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ISSN: 2349-9028
Volume 9, Issue 2, 2022
DOI (Journal): 10.37591/JoMTRA

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Review JoMTRA

Hawk Eye Detection System (Using Python)

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Abstract

Tennis and cricket are gaining popularity. Hawk eye is a modern tool that can be applied to any sport. Now a days there is lots of advancement in the sports technology. Hawk Eye first gained notoriety in television broadcasts of cricket. Broadcasters used Hawk-Cricket Eye's System at the 2006 ICC Champions Trophy in India, the World Cup, and important Test and ODI series all over the world since 2001. A virtual reality device called Hawk-Eye is a line calling system that tracks a ball's trajectory. It has become common due to the plethora of help this technology offers to enhance game quality. Six calibrated high-speed professional vision processing cameras that are distributed throughout the ground are used by Hawk-Eye. Additionally, the system calibrates the two "Mat" broadcast cameras it utilises to ensure that the visual is always superimposed in the proper location. The way in which it operates then allows the game's officials to understand where the ball has pitched and what its onward trajectory would have been. The Hawk-Eye framework is divided into two sizable components: the Tracking System, which includes a camera and a speed weapon, and the Video Replay System, which records data to aid arbitrators in making decisions and to aid coaches and players in analysing their performance. However, there is very less research available in this area. Examples of Hawk-Eye statistics include: LBWs, Wagon Wheels, De Spin, Pitch Maps, Beehives, RailCam, Reaction Time, Ball Speeds. The purpose of this paper is to discuss the principle behind Hawk Eye, many of its applications, future developments, system precision, and system dependability in the sport of cricket.

Keywords: Hawk Eye, DRS, LBW, Despin, cricket, tennis, camera technology

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Received Date: July 18, 2022
Accepted Date: August 06, 2022
Published Date: August 18, 2022

Citation: M.S. Bewoor, Sheetal S. Patil, Avinash M. Pawar, Abhishek Kumar, Akash Sharma, Aryan Sharma, Ashmanshu Nepal. Hawk Eye Detection System (Using Python). Journal of Multimedia Technology & Recent Advancements. 2022; 9(2): 8–14p.

INTRODUCTION

Sports broadcasts from a popular location. As a result, streaming channels are competing fiercely to become the leading candidate. To accomplish this, the channel aims to provide viewers with the best possible experience by utilising cutting-edge technology. Another technology used in broadcasting production channels for a variety of visuals, such as coloured waggon wheels, is Hawk-Eye.

Players can now play games and watch slow-motion videos. As a result, they might decide inadvertently according to the umpire, which would mean that the final result of the game would be different. As a result, it makes sense that the umpire's call would be less precise if the ball travels faster than 90 mph. The DRS system has been enhanced as a result, enabling players to

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question the umpire's judgement. Players can now play games and watch slow-motion videos. As a result, they might decide inadvertently according to the umpire, which would mean that the final result of the game would be different. As a result, it makes sense that the umpire's call would be less precise if the ball travels faster than 90 mph. The DRS system has been enhanced as a result, enabling players to question the umpire's judgement

Now, players may enjoy both gaming and slow-motion content. As a result, they can unintentionally choose in accordance with the umpire, which would change the game's outcome. Therefore, it makes obvious that if the ball is travelling faster than 90 mph, the umpire's call would be less accurate. As a result, the DRS system has been improved, enabling players to contest the umpire's decision.

Even though Hawk-eye technology is frequently utilised in cricket, little research has been done on how generally known it is. In order to raise public knowledge of Hawk-eye technology, this study's goal is to do so. In this study, the term "hawk-eye technology" is defined, and its applications to cricket are covered. This study will concentrate on the reliability and correctness of the current system before taking into account any adjustments in the future. In order to acquire data and statistics from ESPN, this study will make reference to already published sources. A study of Hawk-effects eye's on cricket will be published on the Cricinfo website.

Experimental-section

This section describes studies related to the following categories [1]:

1. Hawk Eye Technology
2. Wagon Wheel
3. The LBW Decision
4. Despin Graphics
5. Beehive Graphics
6. Pitch Map
7. Uncertainty of the Hawkeye System
8. Dependability of the Hawk Eye Technology.

Hawk Eye Technology

Following that, these results are used to select the filter pixels that are available for each frame then examine those pixels. Following the analytical step, several representations of some selected pixels were created. These contributions might include renderings of the cricket field, the boundary of the play area, and other important details. Figure 1 can help you comprehend the procedure mentioned above more clearly.

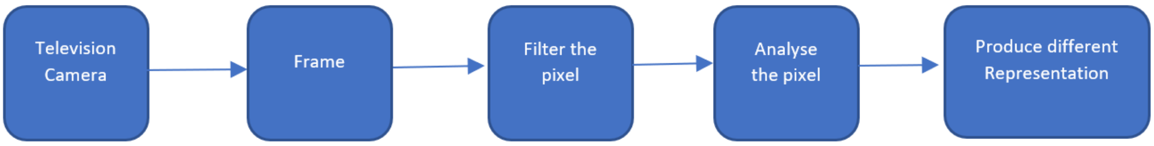

Figure 1. Renderings of the cricket field.

The coordinates of the pixels are given numerically, including their time and home coordinates [2]. During this time, statistical formulas are being employed to recreate the flight of the cricket equipment, the purpose of its impact, and some other significant statistics connected to the area of ground. These data will also account for the specifics relating to completely different pixels present in several frames. After executing the extra calculations mentioned above, the system has the option of leaning.

The foundation of almost all technical systems is a presumption. This Hawk is based on the triangulation hypothesis. It is based on triangulation. In the below figure we can see better

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understanding of the triangulation principal. Which say's by using the given points the angle must recognise the situation which is being measured.

Wagon Wheel

Another cricket-related application of Hawk-eye technology is the manufacture of various types of waggon wheels. These wagon wheels display various data in relation to the shots that a slugger faces. To achieve this the device point out the ball path after the ball is thrown.

Generally, broadcasting channels have used these statistics to improve viewer comprehension. The Figure 2 shows the graphical representation of how the system make the graphics by using the data of sixes, fours, ones, played by the batsmen.

Pundits utilise these images to provide their thoughts on the choice of shot. The Hawk Eye system may also be considered as a helpful tool when playing cricket because players typically use this information to analyse their weaknesses and enhance their skills [3].

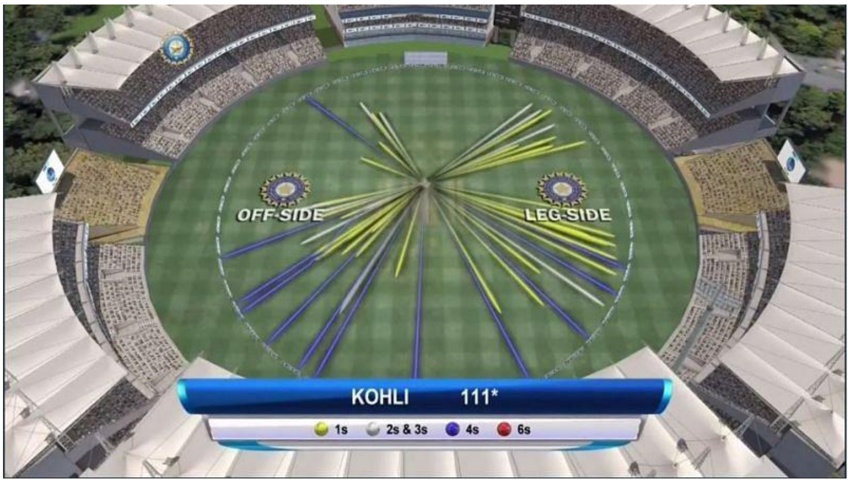

Figure 2. Graphical representation of sixes, fours, ones, played by the batsmen.

LBW Decision

THE LBW DECISION in the game of cricket Associate in Nursing umpire plays a very important role. So, we've got to simply accept that it's terribly tough to form judgments throughout a cricket Ernst Mach [4]. One in every of the foremost crucial selections created by the on-field umpire is that the LBW call. When a batsmen try to or accidentally stops the ball that will hit the legs pads this rule will declare the batsmen is out or not by using lbw call (Figure 3).

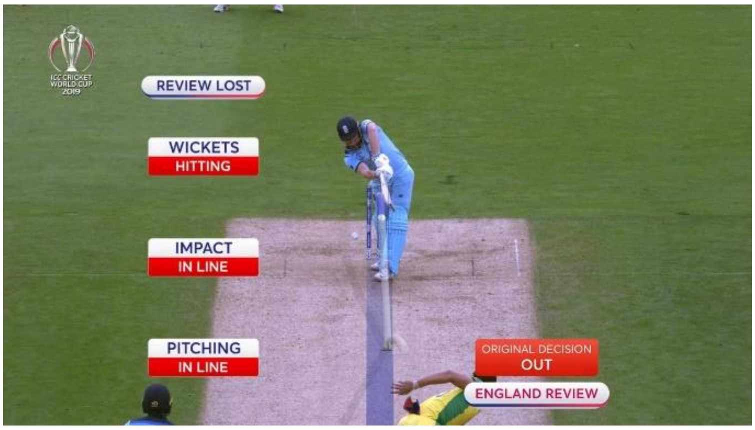

Figure 3. Decision review system.

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Otherwise, the stumps. As a result, it is evident that the on-field umpire needs to decide right away. As a result, there is a possibility that you will make a poor decision that will affect how the game turns out. To address this, the Decision Review System (DRS) was created. The Hawk Eye system is an essential part of the DRS system [5].

Despin Graphics

When looking for a match, despin may be a rather prevalent style of graphic that is viewed.

Generally hawk eye make a virtually representation of the ball. This is done with the intention of watching how the ball moves as it is tossed on the ground.

A batter can improve his game and avoid being outplayed by the bowlers by identifying the strategies used by various bowlers by analysing these visuals. With the aid of such images, it is possible to caterpillar-track not just the types of ball delivered by the bowler.

Beehive Graphics

The most popular type of visual representation is used to show where the cricket ball is in the game of cricket. The slugger's plane serves as the reference point for setting the position. The system leaves its imprint by utilising data on the mechanical behaviour of the cricket equipment. This symbol, which is positioned on the batsman's plane, might highlight where the cricket equipment has passed or will cross the slugger's far side.

Where the slugger has not scored in another colour. As a result, it's much easier to comprehend all of a slugger's strengths and weaknesses. Additionally, the technology has the ability to display an annotation alongside the deliveries that are coloured according to the slugger's score. While concentrating on these graphics, a bowler will target a slugger's weak spots in order to eliminate him while not making a significant impact on the board.

Uncertainty of the Hawk Eye system

Nearly every system in existence is not completely accurate. A little amount of mistake is present in the majority of technical systems [6]. Hawk Eye systems have a tendency to make mistakes occasionally. This occurs as a result of several aspects that may produce a control to the mechanical phenomenon of the ball not being taken into account. The amount of spin a bowler provides, as well as atmospheric elements (wind speed, pitch condition, and ball flight) are not taken into account by the Hawk Eye system [7]. It's crucial to lessen the impact of the Hawk's mistake because the decisions made during a game of cricket can directly affect the context of the sport.

Pitch Map

It is simple to analyze that the cricket pitch was separated into several distinct zones by looking at Figure 3. These areas were divided to support the batsman's space. Utilizing this image, it is possible to see the specific locations where the cricket bowler pitches his or her equipment while the game is in progress.

One can gain insight into the various bowling strategies used by different bowlers by analysing these visuals. These strategies vary depending on the opponents they face, the type of pitch (slow tracks, quick tracks), the variety of weather factors (wind, humidity, sun light), and many other circumstances.

Additionally, it enables the baseball player to see how they performed in comparison to various bowler types and deliveries of various lengths. Therefore, despite the fact that they were initially used in the field of broadcasting, these kinds of graphics can also be viewed as a coaching tool for cricket players. This has contributed significantly to the tremendous advancement of cricket during the past ten years.

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Dependability of Hawk Eye Technology

Data on DRS selections made during cricket matches was acquired from the ESPN cricket info website in attempt to identify the trustworthy Ness of this system, which is being applied to perform the DRS system. The website ESPN Cricket info is taken into consideration as the top cricket website among cricket enthusiasts worldwide for the past 20 years. Consequently, the data acquired from this website is accurate as shown in Table 1.

Table 1. The DRS selections made year by year [5].

Year	Matches	% Overturned	Referral	Overturns
2015	18	30	7.4	2.2
2016	23	28	9.3	2.7
2017	25	24	10.4	2.5
2018	34	21	9.2	1.9
2019	31	24	9.7	2.7
2020	32	32	12.6	4.0

Table 1 presents some statistics on DRS decisions made during check matches between 2009 and 2016. It's important to note that these data don't include situations during which When this technology was first established, it was only employed if the groups demanded that it be put into effect. As a result, only a small number of tournament were organised at the beginning by using DRS [8].

The below statistics were tournament played with DRS Figure Seventeen can be seen in Figure Eighteen's bar chart. This bar graph illustrates the increased trending tendency for associate's degrees in nursing throughout time. So it will be concluded that the system's dependability compelled the game to adopt this strategy. As a result, DRS is currently used in almost all cricket matches.

Table 2 shows some statistics of using DRS the year of 2015 to 2020. The statistics include data on the success rates attained by the look at playing cricket teams in relation to their reviews. In the meanwhile, it will also provide details on each team's bowling and batting success rates.

Table 2. DRS Success Rate (%of Umpire Decision Overturned) 2009–2019 [5].

Team	Success %	Batting %	Bowling %
India	30	60	18
Australia	29	36	24
England	27	34	23
New Zealand	25	36	19
South Africa	28	36	23
Pakistan	23	29	19
Bangladesh	27	43	16

The term Hawk Eye has been consistently associated with the game of cricket for more than ten years [9, 10]. Despite not being a particularly exciting aspect of the game, it quickly gained widespread acceptance in the cricketing community. Even if the Hawk Eye technology had been widely used, public awareness of it is still exceedingly low. While exploring the technology underlying the Hawk Eye, the numerous ways that it is employed in the game of cricket, and the ambiguity of the system, the entire study was undertaken in an effort to improve public comprehension of technology (Figure 4) [5].

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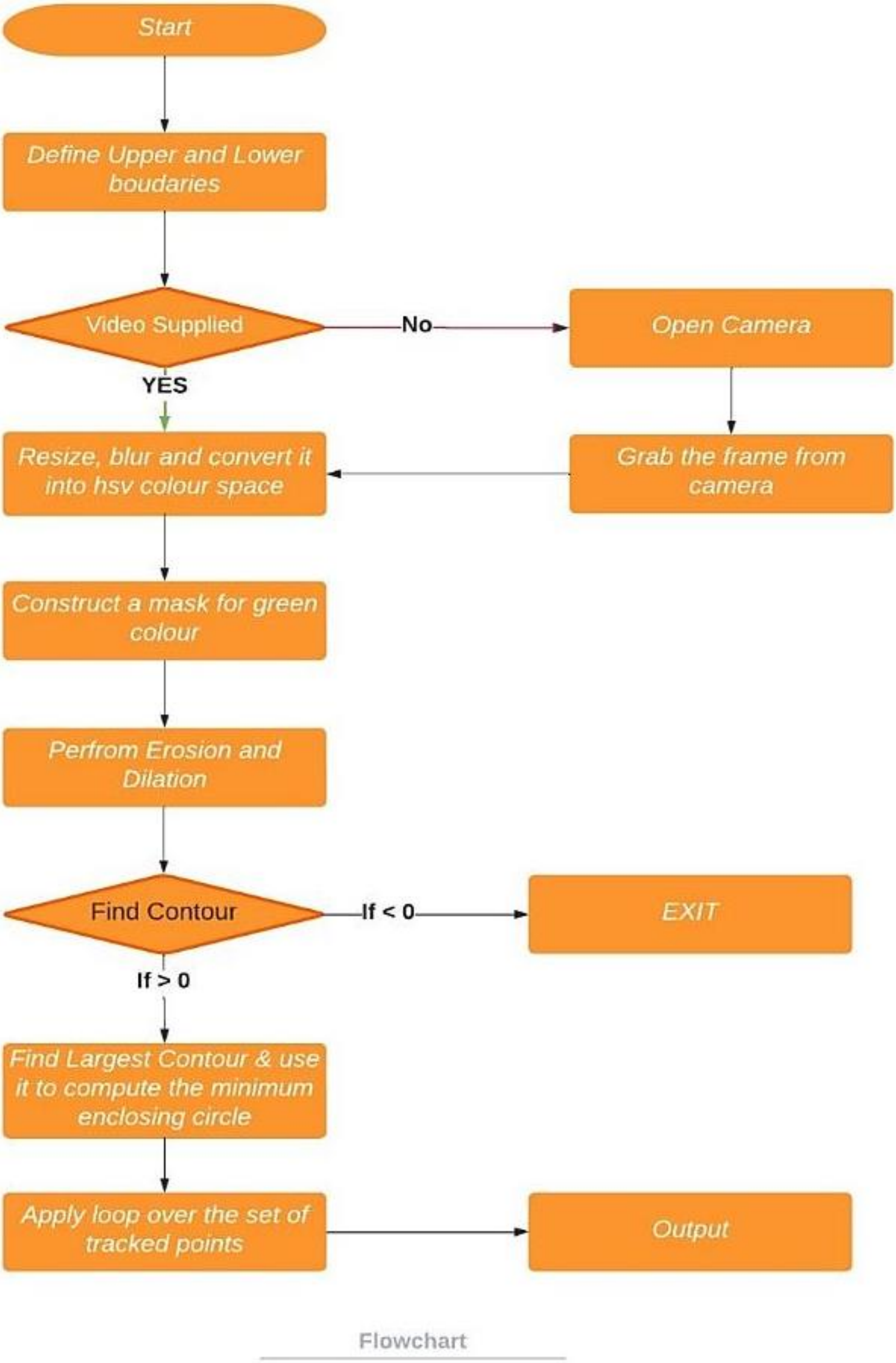

Figure 4. Flowchart of Hawk Eye technology.

CONCLUSION

Quantity was dependent on technology for the game of cricket and its ensuing improvements. The technology's major goal is to study the ball's mechanical behaviour. The video input from cameras is

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gathered in the procedure of tracing the starting trajectory. While processing the video input, the ball's speed will be taken into consideration.

The main 3D position of the ball is decided with the use of "triangulation." The final mechanical phenomena of the ball will be confirmed using software that employs applied mathematics methods while taking into account the mechanical phenomenon determined by the video inputs and expected mechanical phenomenon. The knowledge we get mechanical behaviour of the ball is apply to make meaningful data.

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